SAGAR KESHAVE (AI/ML Engineer)

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EXPERIENCE

Datannovite Llp (Experience - 1 Year)

Role : AI ML Engineer

Code Developments :

- **LLM-powered campaign optimization & generation. (RAG)** June 2024 - Aug 2024
  1. Developed an LLM-powered solution using Retrieval Augmented Generation (RAG) dataset sizes handled(up to 2 lakhs transactions) to analyze client transaction and campaign data, enabling data-driven decision-making and automated campaign generation. **Achieved 100 % accuracy in docs retrieval from vector store (Milvus).**
  2. Improved campaign effectiveness by leveraging RAG-based insights to inform campaign targeting and messaging.
  3. Tech Stack: Python | LangChain | RAG | Milvus | Docker
- **Product Segmentation & Sales Forecasting:** Aug 024 - Nov 2024
  1. Executed ABC-XYZ analysis to segment products by consumption value and demand stability. Built LSTM-based models to forecast sales and quantity with 10% to 15% deviation. Domains - Jewellery, Apparel & clothing, Watch (+4)
  2. Enabling better inventory and promotional planning. Processed millions of transactional records, handling missing, inconsistent data through automated pipelines.

- **DataRobo - RAG** Nov 2024 - April 2025
  1. Addressed the challenge of efficient information retrieval from diverse document formats (KYC forms, PAN cards, invoices, legal contracts) by developing a robust question-answering module.
  2. Overcame limitations in querying large PDFs (size 1000 pages) by optimizing extraction (Llama parser) and processing techniques (Milvus Retriever), resulting in improvement of 30% reduction in processing time.
  3. Integrated text and image data from PDFs into a unified querying system enabling more comprehensive and accurate responses.
  4. Tech Stack: Python, LangChain, RAG, Milvus, Gemini 2.0-Flash, LlamaParser, MongoDB

PROJECTS

- **AI assistant powered by LangGraph (Github)** Sept - 2025
  1. **Worker Node**: A user's request and success criteria are sent to the worker with access to tools. It continually works on the problem, generating a response or deciding to use a tool to gather more information.
  2. **Tools Node**: This node executes the tools the worker has requested, such as browsing a website, running Python code or file handling(edit/write). The output of the tools is then passed back to the worker node for further processing.
  3. **Evaluator Node**: The Evaluator checks if the response meets the **success criteria**. If the criteria are met or if the worker needs clarification from the user, the process ends. If the

criteria are **not met**, the Evaluator sends **feedback** back to the Worker. Evaluator outputs structured feedback via a Pydantic schema (feedback, success criteria met, user input needed)

4. **Edges:** Conditional edges route between worker → tools → evaluator → worker until the task is complete.

- **Image captioning** (Github) Dec 2023
  1. Built an image captioning system addressing the challenge of automatically generating descriptive text for images.
  2. This involved integrating a pre-trained Inception-V3 model for image feature extraction, an LSTM-based decoder with visual attention for sequence generation.
  3. Word2Vec embeddings (initialized with the GoogleNews vectors corpus) to capture semantic relationships between words.

- **Neural machine Translation** (Github) Nov 2023
  1. Built an English-to-Marathi neural machine translation system, addressing the complexities of word-level translation.
  2. This involved training an encoder-decoder LSTM model, incorporating bidirectional LSTMs to capture long-range dependencies.
  3. Integrated Word2Vec word embeddings, refined using the GoogleNews vectors corpus, to enhance semantic representation and caption quality. Bahdanau attention to effectively handling word alignment during translation.

TECHNICAL SKILLS

Programming language - Python
Database - MySQL , SSMS, MongoDB
Cloud - Google cloud platform
Web Framework - Django, Flask, streamlit
Tools and libraries - Tensor flow, NLTK, scikit-learn, langchain, llama-index, beautiful soup, github, streamlit, Google colab, spacy, textBlob, HTML, kaggle, HuggingFace ,

EDUCATION

L.T.D.P senior college, Pune University, Bachelor of computer science, , April 2025
Wingsss college of aviation technology, Pune, Aircraft Maintenance Engineering (Mech), 2017-2021
Skilledge Course Data Science bootcamp, Pune Nov - 2022